

DISTRIBUTED COMPUTING

IV B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS22	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. Understand the foundations of distributed systems.
2. Learn issues related to clock Synchronization and the need for global state in distributed systems
3. Learn distributed mutual exclusion and deadlock detection algorithms
4. Understand the significance of agreement, fault tolerance and recovery protocols in Distributed Systems
5. Learn the characteristics of peer-to-peer and distributed shared memory systems

COURSE OUTCOMES

By the end of the course, the student will able to:

1. Elucidate the foundations and issues of distributed systems
2. Illustrate various synchronization issues and global state for distributed systems.
3. Illustrate Mutual Exclusion and Deadlock detection algorithms in distributed systems
4. Describe the agreement protocols and fault tolerance mechanisms in distributed systems
5. Describe the features of peer-to-peer and distributed shared memory systems.

UNIT - I	A MODEL OF DISTRIBUTED COMPUTATIONS	CLASSES: 12
Definition, Relation to computer system components, Motivation, Relation to parallel systems, Message-passing systems versus shared memory systems, Primitives for distributed communication, Synchronous versus asynchronous executions, Design issues and challenges. A model of distributed computations: A distributed program, A model of distributed executions, Models of communication networks, Global state, Cuts, Past and future cones of an event, Models of process communications. Logical Time: A framework for a system of logical clocks, Scalar time, Vector time, Physical clock synchronization: NTP.		
UNIT - II	MESSAGE ORDERING & SNAPSHOTS	CLASSES: 12
Global state and snapshot recording algorithms: Introduction, System model and definitions, Snapshot algorithms for FIFO channels, Snapshot algorithms for non-FIFO channels, Message Ordering & Snapshots: Message ordering and group communication: Message ordering paradigms, Asynchronous execution with synchronous communication, Synchronous program order on an asynchronous system, Group communication, Causal order (CO), Total order.		
UNIT - III	DISTRIBUTED MUTEX& DEADLOCK	CLASSES: 12
Distributed Mutex& Deadlock: Distributed mutual exclusion algorithms: Introduction, Preliminaries, Lamport's algorithm, Ricart-Agrawala algorithm, Maekawa's algorithm, Suzuki-Kasami's broadcast algorithm. Deadlock detection in distributed systems: Introduction, System model, Preliminaries, Models of deadlocks, Knapp's classification, Algorithms for the single resource model, the AND model and the OR model.		

UNIT - IV	RECOVERY & CONSENSUS	CLASSES: 12
Recovery & Consensus: Check pointing and rollback recovery: Introduction, Background and definitions, Issues in failure recovery, Checkpoint-based recovery, Log-based rollback recovery, Coordinated check pointing algorithm, Algorithm for asynchronous check point in grand recovery. Consensus and agreement algorithms: Problem definition, Overview of results, Agreement in a failure, free system, Agreement in synchronous systems with failures.		
UNIT - V	PEER-TO-PEER COMPUTING AND OVERLAY GRAPHS	CLASSES: 12
Peer-to-peer computing and overlay graphs: Introduction, Data indexing and overlays, Chord–Content addressable networks, Tapestry. Distributed shared memory: Abstraction and advantages, Memory consistency models, Shared memory Mutual Exclusion.		
TEXT BOOKS		
1. Distributed computing: Principles, algorithms, and systems, Ajay Kshemkalyani and MukeshSinghal, Cambridge University Press, 2011. 2. Distributed Systems Concepts and Design, George Coulouris, Jean Dollimore and Tim Kindberg, Fifth Edition, Pearson Education, 2012.		
REFERENCE BOOKS		
1. Distributed Operating Systems: Concepts and Design, Pradeep K Sinha, Prentice Hall of India, 2007. 2. Advanced concepts in operating systems. MukeshSinghal and Niranjana G. Shivaratri, McGraw-Hill, 1994. 3. Java in Distributed Systems, Concurrency, Distribution and Persistence, Marko Boger, 2001. 4. Distributed Systems: Principles and Paradigms, Tanenbaum A.S., Van Steen M., Pearson Education, 2007. 5. Kshemkalyani Ajay D, Mukesh Singhal, Distributed Computing: Principles, Algorithms and Systems, Cambridge Press, 2011 6. Mukesh Singhal, Niranjana G Shivaratri, Advanced Concepts in Operating systems, McGraw Hill Publishers, 1994.		
WEB LINKS		
1. https://onlinecourses.nptel.ac.in/noc21_cs15/preview 2. https://www.distributedsystemscourse.com/		